

Hybrid Deep Learning-based Feature-augmented Detection For Molecular Communication Systems

Fardad Vakilipoor, Davide Scazzoli, Francesca Ratti, Gabriele Scalia, Maurizio Magarini
Politecnico di Milano, Milan, Italy, Email: (name.surname)@polimi.it

ABSTRACT

This manuscript presents a hybrid Deep learning algorithm to detect transmitted symbols from experimental measured data in a biological molecular communication (MC) testbed. The algorithm consists of two deep neural networks capturing different aspects of the time signal, which are combined based on their relative confidence. The MC testbed uses a transfected *Escherichia coli* (*E. coli*) bacteria that express the light-driven proton pump *gloeorhodopsin* from *Gloeobacter violaceus*. Given an external controllable light stimulus, driven from a light-emitting diode (Transmitter), the bacteria secrete protons that change the pH level of the environment. A pH detector (Receiver) measures the pH of the environment. Modelling such a biological real system accurately is not feasible. Thus, in order to detect the transmitted bits we use both a convolutional and a recurrent neural network in tandem. This paper discusses the data augmentation, processing, and neural networks pertinent to a practical MC problem. The trained algorithm detects the transmitted bits with an accuracy above 99.9%.

CCS CONCEPTS

• **Hardware** → **Biology-related information processing**; • **Applied computing** → **Biological networks**.

KEYWORDS

Molecular communication, hybrid deep learning, symbol detection.

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1 INTRODUCTION

Molecular Communication (MC) is a new information and communication paradigm based on mechanisms and materials from biological systems [1]. Owing to its bio compatibility and energy efficiency unlike the classical communication methods, MC finds opportunity to be used as an information transmission paradigm

in sub-millimeter range. Smart drug delivery [2, 3], health monitoring [4], and disease diagnosis [5] are just few examples that MC was able to bring hope in order to close the gap between reality and theory.

As the initial step like any conventional communication studies, researchers have investigated different aspects of MC theoretically using simplifying assumptions [6]. Channel modeling [7, 8], channel capacity [9], detection problems [10], and localization [11] are just part of the theoretical investigations that have been conducted to understand MC systems. It is worth noting that the focus of this work is on symbol detection. Detection is one of the most crucial aspects of any communication system. Linear memory-less detection [10], adaptive detection [12], anti inter-symbol interference (ISI) demodulation [13], and matched filter [14] are some of the theoretical solutions for detection problem in MC literature. They all rely on the knowledge of channel impulse response and are considered as coherent methods. In practical MC systems, looking for a closed-form solution that is capable of describing the phenomena and detect the transmitted symbol is laborious or impossible, due to ISI, noise, non-linearity and time-variation of the channel properties [15]. Thus, machine learning algorithms may be preferable. We train a detector without any knowledge of the underlying channel models. The receiver goes through a training phase where a deep learning detector is trained using known transmission signals. Sharma *et al.* [16] propose a deep neural network (DNN) receiver design with two hidden layers to decode the data symbols in MC. They generated the test and training datasets based on the analytical channel impulse response of a diffusive channel with binary concentration shift keying modulation. Authors in [15] proposed a technique called sliding bidirectional recurrent neural network for sequence detection in channels with memory where, after training, the detector estimates the data in real time as the signal stream arrives at the receiver.

In this work, we design a hybrid deep learning model consisting of two DNNs to detect the transmitted symbols in a biological MC system. Fig. 1 shows the overview of the proposed approach. The MC testbed, which is defined in [17], consists of transfected bacteria that secrete protons in response to light. The released protons change the pH of the medium, which can be measured by a pH sensor. Equivalently, from an input-output perspective, the MC system converts a controllable optical signal from light-emitting diode (LED), into a chemical signal by changing the pH. The algorithm is trained by using simulated data generated from parameters obtained through experiments [17]. The training dataset contains ten realizations of the MC system for 1000 consecutive bits transmitted over the channel, along with the sampled pH detected at the receiver. The symbol interval is one minute, and the pH detector sampling frequency is 1 Hz. Each of the ten realizations is under different and unknown channel conditions.

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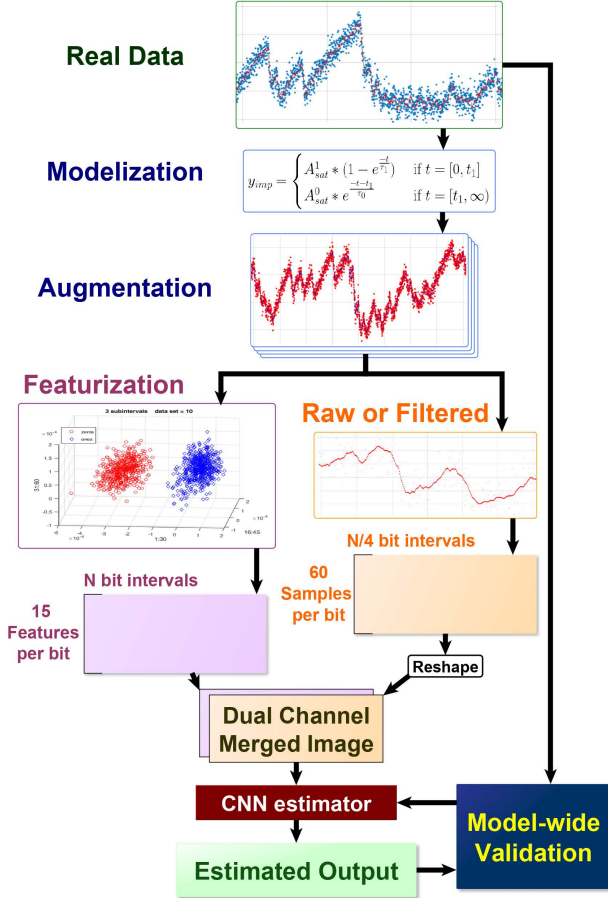


Figure 1: Overview of the approaches described in this paper. The real data is described in Sec.2, the modelization and augmentation are described in Sec.4, the featurization and filtering are described in Sec.3 while the data merging and estimator are described in Sec.5.

The remainder of this paper is structured as follows. Sec. 2, describes the testbed and datasets. In Sec. 3, we elaborate on the feature extractions and data processing algorithms of the received signal. Sec. 4 presents the data augmentation method. Sec. 5 discusses the architecture of the proposed DNNs and demonstrates the bit prediction accuracy. Finally, Sec. 6 discusses results and conclusions.

2 TESTBED AND DATASETS

The training datasets throughout this study had been generated by simulation based on parameters obtained by experiments with real bacteria [18]¹. The testbed is realized using *E. Coli* bacteria that express the light-driven proton pump *gloeorhodopsin*. Bit one corresponds to switching on the LED for a fraction of symbol interval, while bit zero is associated with no light emission to the bacteria culture during the symbol interval. Upon an external light stimulus, these transfected bacteria export protons into the channel.

¹The datasets were published for the ACM NanoCom 2021 data competition [19]

By emitting light the pH of the medium goes down and returns to its steady state otherwise. In this way, a sequence of light pulses can be converted into a pH signal that carries the underlying information. The training dataset contains 10 samples of 1000 consecutive bits transmitted over a molecular communication channel along with the signal detected at the receiver. Each of the 10 samples is under different channel conditions. The test dataset, which will be used to evaluate the performance of the algorithm can also be under different channel conditions. Note that, the testbed is based on the reactions. Thus, we expect that the effect of secreted protons on the pH lasts longer than the corresponding symbol interval. This introduces the notion of memory in the channel and causes severe ISI. Fig. 2 shows the pH variation and the corresponding bits over a short time interval.

3 DATA PROCESSING AND FEATURES

The training dataset shows a steep variation of pH in symbol intervals associated with bit one, Fig. 2. However, due to noise, memory effect, non-linearity, and time varying nature of the model, detecting the true bit from pH signal might be challenging even through human inspection. To featurize the pH signal in a symbol interval, we apply a number of preprocessing steps. Moreover, we analyze the response in the frequency domain and design a low pass filter. The preprocessing methods are described in the following.

3.1 Polynomial Curve Fitting

The ISI effect in the channel introduces correlation between the observed signals in time domain. To extract the features of the observed signal in a symbol interval, we hired polynomial curve fitting method. We did this operation before removing the noise. Because we believed that passing the data through a low-pass filter may destroy some correct information as it has been proven by the data processing theorem [20]. We considered the features as the coefficients of the k^{th} order polynomial fitted on the pH curve during each symbol interval. To evaluate the helpfulness of the extracted features, we defined a space called feature-space and each dimension of the space represents the value of a polynomial's coefficient. Fig. 3 shows the feature-space of dataset number 10, where each dimension is the coefficients of the second order fitted polynomial within a symbol interval. In this particular case we can observe that both clouds of features corresponding to bits one and zero are separable. However, not all datasets three dimension (3D) feature-space are easily separable into two distinguishable clouds. Fig. 4 shows the feature-space of dataset number 5. In this case, we can observe that the clouds of bits are not easily separable in 3D space with the aforementioned dimensions. As a preliminary analysis the observations from Figs. 3 and 4 show that the polynomial coefficients are reliable features.

3.2 Piece-wise derivatives

As shown in [18], the relation between bit one and pH might be observed by analyzing the derivative of pH signal. However, due to the previously mentioned effects, describing the whole symbol interval with only one value as the derivative could be misleading. Thus, we break a symbol interval into multiple sub-intervals and compute the derivative that is equivalent to the slope of the first

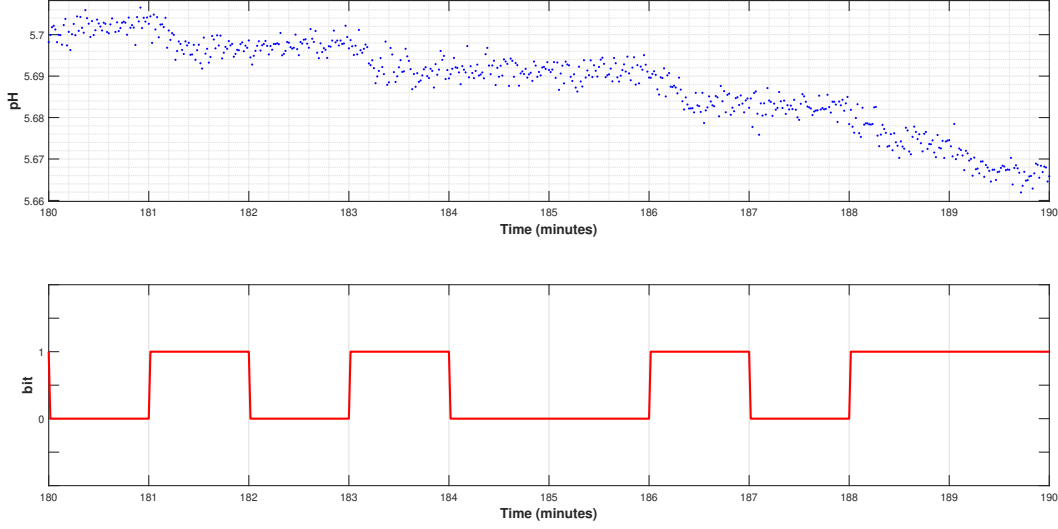


Figure 2: pH samples and their corresponding bits from dataset 6 within time interval 180 to 190 seconds

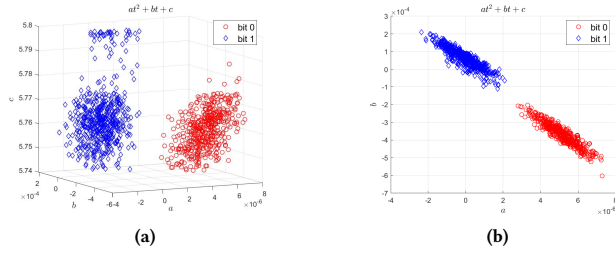


Figure 3: (a) Feature-space of dataset number 10, where each dimension stands for the coefficients of the 2nd order fitted polynomial. Blue diamonds correspond to bit one and red circles correspond to bit zero. (b) Demonstrates the feature-space from a 2D perspective.

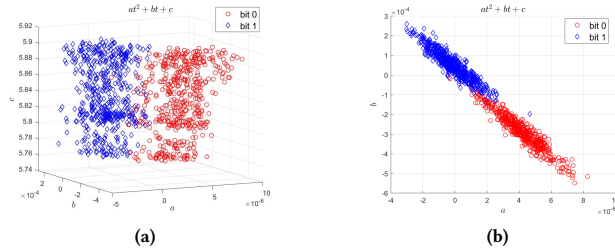


Figure 4: (a) Feature-space of dataset number 5, where each dimension stands for the coefficients of the 2nd order fitted polynomial. Blue diamonds correspond to bit one and red circles correspond to bit zero. (b) Demonstrates the feature-space from a 2D perspective.

order line fitted in each sub-interval. The problem here is, introducing multiple sub-intervals reduces the number of samples in each of them. Thus, the feature, which is the slope of the line in this case become sensitive to noise and other distortions. To avoid this problem We introduce an overlap between sub-intervals to include more samples and ensure that slopes are robust against noise. Fig. 5 demonstrates the 3D feature-space, as each dimension represents a slope of sub-intervals 1 to 30, 16 to 45, and 31 to 60 seconds. Note that each interval was 60 seconds. An interesting observation is that both clouds form spread approximately spherically for all datasets. From this figure one can observe that the points are scattered in all dimensions of the feature-space without any alignment. This spherical morphology of clouds suggests the independence between the features, which are the dimension of the space. From this observation we conclude that there is no repetition of information in these sets of features to feed to the DNN.

3.3 Filtering

From the analysis in the frequency domain of the training data, it can be noticed that the noise can be approximated as an additive white Gaussian noise. Therefore, it is possible to design a low pass filter that attenuates the high frequency components. Due to the biological testbed's slow nature, high frequency components are mostly just noise. After removing high frequency components, we apply Savitzky-Golay smoothing [21].

4 DATA AUGMENTATION

The limitation of insufficient data to use for training neural networks is a well understood problem, several solutions are being presented in the literature, with a very promising concept being the generation of synthetic data to train a neural network [22]. We design a model to generate new synthetic realistic data to be used

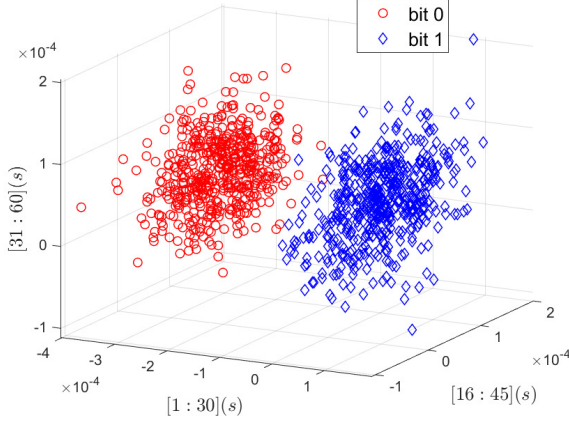


Figure 5: Feature-space of dataset number 8, where each dimension stands for the derivative of sub-intervals 1 to 30, 16 to 45, and 31 to 60 seconds. Blue diamonds correspond to bit one and red circles correspond to bit zero.

downstream. One of the ways in which the data can be modeled is shown in [18], where we can extract an equation to model the channel impulse response as shown in the following

$$y_{imp}(t) = \begin{cases} A_{sat}^1 (1 - \exp(-\frac{t}{\tau_1})) & \text{if } t = [0, t_1] \\ A_{sat}^0 \exp(-\frac{t+t_1}{\tau_0}) & \text{if } t = [t_1, \infty) \end{cases} \quad (1)$$

As the quality of the prediction is heavily dependent on the quality of the synthetic data generated, we improve this model by introducing a saturation variable. This variable interacts with the impulse generation parameters, $A_{sat}^1, A_{sat}^0, \tau_1, \tau_0$ by reducing its amplitude and duration linearly depending on the saturation level, introducing behaviours in the synthetic data which have been observed in the real data. The effect introduced is calculated by a weighted average of three possible causes, which were identified in the provided data:

- **Current pH level:** as the global pH tends to get lower then it is reasonable to assume the medium becomes saturated and further lowering the pH becomes harder.
- **Current pH trend:** when the pH is transitioning to a low level we have observed from the data that it becomes harder to distinguish transmitted bits, which become hidden inside the overall trend of the pH value.
- **Previously transmitted bits:** since the transmission of a positive bit involves the release of molecules that reduce pH, if a sequence of positive bits is transmitted, the bits at the end are likely to be less prominent as the medium becomes saturated by the molecules released by earlier bits.

The prominence of each of these effects is mediated by a weight which is randomly assigned when creating a new channel. All weights and parameters for both unsaturated and fully saturated conditions are initialized randomly, within an interval of values which allow for the generation of synthetic data that closely matches the real data provided [17]. The overall pH trend observed in the real data contained both linear realizations and sinusoidal ones, an

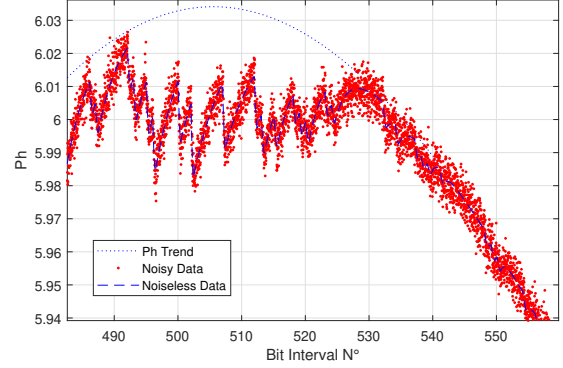


Figure 6: Example of augmented data generated, the effect of saturation from trend can be seen on the right

example of the latter is shown in Fig.6. Both have been replicated in the synthetic data, furthermore when the trend was sinusoidal, trend based saturation was more prominent, this effect was easily included by tuning the corresponding weight.

5 PREDICTORS

5.1 Recurrent Neural Network

To model the real data, we compare different architectures and pre-processing strategies. The evaluation strategy consists of a 10-fold cross validation, where the realizations of the system are split as 8/1/1 for the train, validation and test set. This ensures that data from the same realization are never used both for training and testing. The evaluated architectures include Convolutional Neural Networks (CNNs) with temporal sliding windows and Recurrent Neural Networks (RNNs). Among them, the architecture reporting best result in a preliminary analysis is an RNN, and in particular a Gated Recurrent Unit (GRU) [23]. We notice that similar architectures have recently reported success in related applications [15]. In terms of features, different combinations of preprocessing strategies described in Sec. 3 have been grid-searched using the evaluation strategy aforementioned. Best results have been obtained using polynomial curve fitting up to degree $k = 3$ (excluding bias), piecewise derivatives ($num_subintervals = 3$) and frequency filtering. Moreover, best results have been obtained using $L = 3$ recurrent layers and $T = 3$ intervals before and after the target symbol interval.

5.2 Convolutional Neural Network

Given the spatial relationship and intrinsic meaning associated to the samples in our input matrix a CNN could be a good fit for a predictor. We built a CNN algorithm through a shallow neural network with a convolutional layer and a fully connected layer, as shown in Fig. 7, with a convolutional layer and a fully connected layer of variable size. The convolutional layer is characterized by M filters of variable size depending on the number of bits considered. By using all the features described in Sec. 3, we obtain 15 features per bit which are fed to the CNN alongside the real data after it is reshaped to match the input dimensions. To separate between

Table 1: Grid search optimization for hyper parameters N and M

		N						
		4	8	16	32	64	128	256
M	4	99.6033	99.7733	99.5900	99.6567	99.7133	99.8100	99.8100
	8	99.7467	99.7433	99.7833	99.7967	99.7967	99.8167	99.8167
	16	99.7200	99.7567	99.7967	99.8267	99.7967	99.7967	99.8100
	32	99.7600	99.7800	99.7867	99.8067	99.8033	99.8133	99.8067
	64	99.7967	99.7933	99.7833	99.7800	99.8033	99.7967	99.7967
	128	99.8000	99.7800	99.8000	99.7933	99.7767	99.8000	99.7900
	256	99.7700	99.7867	99.7967	99.7867	99.8000	99.7967	99.7800

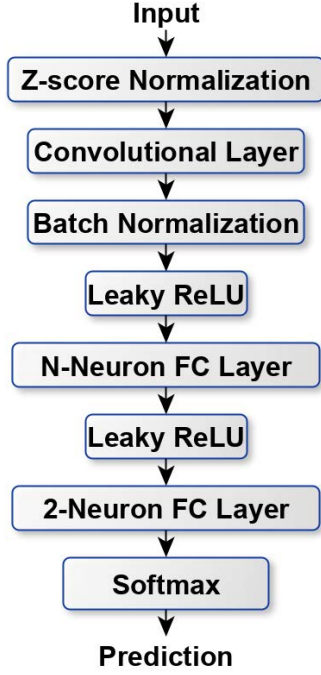


Figure 7: Structure of the CNN predictor. The input is of size $15 \times FL$, where FL is the featurized length which is the number of bits used to create features. The first convolutional layer is characterized by M convolutional filters of size 15×1 . The second fully connected layer is characterized by N Neurons, with the last layer giving the estimated bit output.

the two, a dual channel image is built where 60 bits are described through 15 features and put into a $15 \times FL$ image alongside the full 60 samples of $FL/4$ bits which altogether result in a $15 \times FL \times 2$ input for the CNN. The temporal window of FL bits, where FL is the featurized length which is the number of bits used to create features, for the feature image is divided in past, present and future bits. All of this data contribute to the estimation of the present bit only. This approach emphasizes the importance of close bits while still providing a broader vision through the features. The number of filters for the convolutional layer, M and the number of neurons for the fully connected layer N have been determined through a process of grid

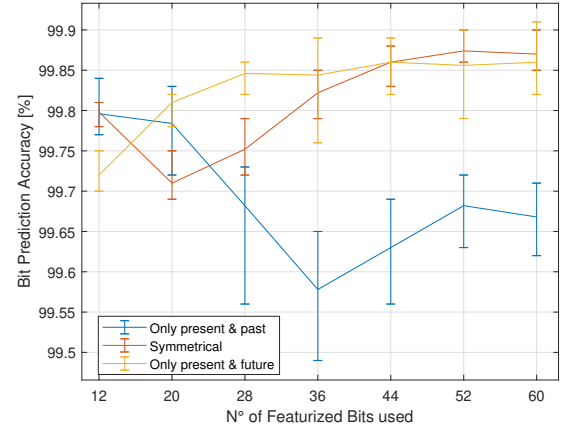


Figure 8: Results of input size optimization. Error bars show min and max value obtained through 5 iterations.

search over the values $N = M = [4, 8, 16, 32, 64, 128, 256]$, where best performance was obtained for the combination $M = 16, N = 32$ although there was not a large difference for other combinations, as is shown by the results reported in Tab. 1. Afterwards the optimization veered towards finding how many bits should be fed to the CNN and whether past or future bits contained more information. We studied three configurations: only past and present bit, only future and present bit and symmetrical. The CNN was built and trained in MATLAB, using Stochastic Gradient Descent with Momentum (SGDM) training algorithm with its default options, with the exception of batch size that was chosen as large as possible while still fitting inside the memory constraints of the GPU. The results obtained from this are shown in Fig. 8. We have observed that for small number of bits symmetrical and past approach outperform the approach based on future bits however, as the number of bits increases the performance of the past approach quickly degrades while that based on future bits only reaches similar performance with the symmetrical case. This suggests that there is information in future bits that is useful to the detection of the present bit.

5.3 Decision maker

The final decision on the received symbol is taken by combining the predictions of the two introduced networks. Both models output a

prediction on the transmitted symbol y_i and an uncertainty on its accuracy σ_i^2 , where $i = 1, 2$, indicating the results of the CNN and of the RNN, respectively. Note that we define $\sigma_i^2 = \min((1 - y_i), y_i)$.

The decision maker determines final estimation of the transmitted symbol $\hat{y}$ by taking the average of y_1 and y_2 weighted for their uncertainties. In formula, it becomes $\hat{y} = (y_1\sigma_1^2 + y_2\sigma_2^2)/(\sigma_1^2 + \sigma_2^2)$. In this way, for each transmitted symbol, the model which provides the less uncertain prediction is privileged in the determination of $\hat{y}$.

6 DISCUSSION AND CONCLUSION

The CNN approach, using 16 filters and 32 neurons for the first and second layer respectively, trained and validated on the synthetic data alone, was able to reach an accuracy of almost 99.9% on average on the provided test data. Our study for the optimization of the CNN input size highlighted the odd behaviour in which future bits carry useful information in the estimation of the current transmitted bit. Using the real data with the RNN (with the hidden size of 8) we obtained an average accuracy of 99.7% on all realizations (using 8 realizations for training and 1 for test, in turn). An analysis of the results highlights that the errors of the two networks are often not aligned, thus strengthening the need of a hybrid approach. Overall, the final model achieves 100% accuracy on many realizations. Further development and extension of this algorithm (including its application to other MC systems) will be the subject of future work.

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